

Oxide deposition – Minutes of the Meeting with Jiří Bulíř

Thur. 04.08.2016

We met Jiří Bulíř from the Department of Analysis of Functional Materials from the Institute of Physics of the AS CR, in Slovanka, Prag. The goal was to know the feasibility of the deposition of oxides of controlled thicknesses between 10 and 200 nm, on metals Ti and Al. More precisely, the systems Ti/TiO₂ and Al/Al₂O₃ are of interest to study the HSF theory developed by Thibaut J.-Y. Derrien. Duration of the meeting: 1h.

His prof. website: <http://www.fzu.cz/~bulir/>

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1 The Meeting

What happened?

- We exposed what we would like regarding oxides and thickness
- We exposed the context of the LIPSS and HSFL and the possibility of tuning their spatial period with the oxide layer
- It was then decided what we can start to do
- More questions
- JB showed us his deposition lab

What devices do they have?

They have a lot of devices for depositing layers, making multilayers, controlling, in real-time (second time scale), the thickness evolution, characterising the roughness and the properties

- magnetron sputtering
- PLD
- electron beam sputtering

Is the guy interested?

- It seems that he was interested! He didn't new about LIPSS but he did about plasmons
- However, he proposed that before going to the whole targets, we will do only 3 for Al/Al₂O₃ to check if it's already working with these 3 points on the SPP curve and if yes we go further.

2 Practical possibilities

What thicknesses are possible

According to him up to 200 nm should not be a problem neither for TiO_2 nor Al_2O_3 . He will use magnetron sputtering to have better homogeneity of the deposition.

Also at the thicknesses we target, it will not further oxidize if in air (which would increase the thickness without control)

What roughness can he get

- In his depositions, the roughness strongly depends on the roughness of the initial metallic substrate.
- He tells that it should be less than 40 nm and better for thinner films.
- If he needs to heat the substrate (typically for getting polycrystalline instead of amorphous) the roughness will be badder
- When growing Al_2O_3 , some kind of independent columns (or kind of pillars) of grown materials are usually seen. this makes the homogeneity not perfect and add some roughness (the top of the columns can be for examples slightly higher than the rest). This make the Al_2O_3 porous... Sorry i understood the concept but really the details in this.

Can they characterize the thickness?

yes, they do with ellipsometry method and can put in the model of calculation roughness. If i understood well their precision is about $\pm 5\%$ at low thickness (10-50 nm) and it increases for higher thickness.

What purity can they get?

They work hith 10^{-5}Pa vacuum and have very high purity targets, so except the contamination of hydrocarbonates from air before deposition and slightly from the chamber, it should be really high purity.

How to store the sample before irradiation?

- basically in some small low vacuum box
- Al_2O_3 is usally very absorbing water so add dessicative perls in the box

3 So what should we do?

We proposed him first to send him a report explaining roughly what we would like to demonstrate and what to be done for this (typically the paper you started on Overleaf git). Then, if he finds it attractive and agrees, we go for a first demonstration with 3 samples, he can also maybe propose something and then if it is encouraging we go further (though he already seems to be ok for the 3 samples).

Send JB the report of what we target

I told him that we will send him some report on next week. Also that it is not urgent, he goes on holiday soon but don't know when. Note, in this report it could be nice to answer the questions that he asked and was not able to answer:

- Why these native oxides (Al_2O_3 on top of Al and TiO_2 on top of Ti and not others)?
- Why don't you want roughness in the end? But now i realize that it should be to be sure about the interference distance?
- For Al_2O_3 , it crystallizes in different phases with "quite differnt optical properties". Which data did you take for your calculations for Al_2O_3 ?
- For Al_2O_3 , do you want it amorphous, or polycrystalline?

Provide substrates

We don't have a lot of money remaining (only 1300 EUR) and we will not be able to buy a lot, especially if we go for gold on the other experiment campaign. Luckily, for samples of 1x1cm, he seems okay to give us 3 samples of Al/ Al_2O_3 with the desired thickness. He said he has already aluminum. Inam proposed also to bring him the 3-4 Ti samples that he polished. It seems that he said yes, but at the same time he was proposing not to make a lot of samples first. Moreover, although Inam manage to get better roughness than shown in Group meeting, it can still be improved i believe and (see below) roughness of the substrate is critical for the roughness of the oxide surface also. But if we follow these info, then:

- JB provides the Al substrate and grow Al_2O_3 on top of it with 3 different thicknesses at the first stage
- We provide the Ti sbstrate
- If your theory also goes for Si/ SiO_2 , as proposed Inam, we have wafers and can provide Si substrates